

Nauman Khan

Senior Data Engineer | Data Platform Engineer | ETL/ELT Architect

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PROFESSIONAL SUMMARY

Senior Data Engineer with 8 years of experience building AI-ready data platforms, real-time streaming systems, and cloud-native ETL/ELT architectures across AI infrastructure, healthcare analytics, and financial data ecosystems. Proven expertise designing scalable lakehouse platforms and distributed data pipelines using Apache Spark, Databricks, Kafka, Snowflake, and AWS processing millions of daily events with high reliability and low latency. Strong background in healthcare data compliance, real-time financial analytics, machine learning data pipelines, and enterprise-scale data platform modernization. Experienced in data modeling, streaming architectures, cloud infrastructure automation, and observability engineering supporting analytics, AI/ML, and business intelligence workloads.

PROFESSIONAL EXPERIENCE

Mage 01/2022 – Present

Senior Data Engineer — AI & Cloud Infrastructure

- Architected scalable AI-ready data platform using Databricks, Apache Spark, and Kafka supporting 30M+ daily events across analytics and ML workloads
- Designed enterprise lakehouse architecture on AWS S3 and Delta Lake enabling unified batch and real-time processing pipelines
- Built low-latency streaming infrastructure using Kafka and Spark Structured Streaming achieving sub-second data delivery with 99.9% uptime
- Developed feature engineering pipelines supporting downstream machine learning and AI inference systems
- Led migration from legacy ETL systems to cloud-native ELT architecture reducing processing time by 40% and improving data freshness SLAs
- Implemented Infrastructure-as-Code and CI/CD pipelines using Terraform and GitHub Actions enabling automated deployment of distributed data services
- Established enterprise-grade data observability and quality framework reducing downstream production incidents by 35%
- Optimized Databricks and Snowflake workloads lowering cloud compute costs while improving analytics performance

Innovaccer 06/2019 – 12/2021

Data Engineer — Healthcare Analytics Platform

- Engineered HIPAA-compliant healthcare data pipelines processing millions of patient and clinical records across cloud-native analytics systems
- Integrated EHR and FHIR datasets into centralized healthcare lakehouse architecture supporting operational and patient analytics
- Designed scalable Snowflake warehouse models enabling self-service reporting for clinical, finance, and population health teams
- Developed distributed ingestion frameworks integrating claims, patient monitoring, and healthcare event streams into unified data pipelines
- Built real-time healthcare analytics workflows supporting patient care dashboards and operational reporting systems
- Improved Spark and SQL workload performance by 45% across enterprise healthcare reporting environments
- Implemented automated data validation and monitoring systems improving healthcare data reliability and compliance reporting
- Supported secure cloud migration initiatives across AWS and Azure healthcare environments

Prophecy.io 07/2017 – 05/2019

ETL Developer — Financial Data Platforms

- Designed and optimized large-scale ETL pipelines processing financial transaction and payment data for enterprise analytics workloads
- Built real-time transaction ingestion pipelines supporting fraud detection and payment monitoring systems
- Developed reusable SQL-based transformation frameworks standardizing financial reporting across multiple business domains
- Improved query execution performance by 35% through indexing, partitioning, and distributed processing optimization strategies
- Supported migration from legacy ETL systems to cloud-native Spark and Snowflake-based analytics workflows
- Implemented data quality and reconciliation checks improving transaction accuracy and reporting reliability
- Assisted in development of event-driven financial analytics pipelines using Kafka and Spark Structured Streaming
- Collaborated with analytics and risk teams to support near real-time reporting and operational insights

TECHNICAL SKILLS	
Languages: Python, SQL, Scala, Java, Bash	
Big Data & Distributed Systems: Apache Spark, Spark Structured Streaming, Hadoop, Apache Flink, Databricks, Apache Kafka, Delta Lake, Apache Hive, Apache Iceberg, Apache Hudi	
Data Orchestration: Apache Airflow, dbt, Luigi, Prefect	
Cloud Platforms: AWS S3, Redshift, Glue, Lambda, EMR, EC2, Athena, Kinesis, IAM, CloudFormation, GCP BigQuery, Dataflow, Pub/Sub, Azure Data Factory, Synapse	
Databases & Warehousing: Snowflake, PostgreSQL, MySQL, MongoDB, Cassandra, DynamoDB, Amazon Redshift	
Data Engineering Concepts: ETL/ELT Pipelines, Lakehouse Architecture, Medallion Architecture (Bronze/Silver/Gold), Batch & Streaming Systems, Data Modeling, Star Schema, Snowflake Schema, Data Contracts, Schema Evolution, Data Lineage, Master Data Management, Data Governance	
DevOps & Infrastructure: Docker, Kubernetes, Terraform, Helm, Git, CI/CD, GitHub Actions, Jenkins, GitLab CI	
Data Quality & Observability: Great Expectations, Monte Carlo, Data Validation Frameworks, Pipeline Monitoring, Logging & Alerting, CloudWatch, Prometheus, Grafana	
Performance & Optimization: Query Optimization, Partitioning Strategies, Indexing, Distributed Computing Optimization, Workload Tuning, Cloud Cost Optimization	
Streaming & Real-Time Systems: Kafka Connect, Kafka Streams, Event-Driven Architecture, CDC Pipelines, Real-Time Analytics, Change Data Capture	
Healthcare & FinTech Data Domains: HIPAA Compliance, FHIR, EHR/EMR Integration, Financial Transaction Pipelines, Fraud Analytics, Risk Data Processing	
AI & ML Data Engineering: ML Feature Pipelines, Vector Data Pipelines, Feature Engineering, ML Data Preparation, AI-Ready Data Platforms, RAG Data Pipelines	

EDUCATION	
Bachelor of Science in Computer Science	2017

SYSTEM DESIGN PROJECTS	
Real-Time Streaming Data Platform - Designed end-to-end event streaming architecture using Kafka and Spark Structured Streaming - Implemented ELT pipeline storing data in AWS S3 and Snowflake for analytics and reporting workloads - Achieved near real-time processing capabilities supporting operational dashboards	
Enterprise Data Warehouse Modernization - Designed scalable Snowflake-based data warehouse using star schema and ELT optimization patterns - Improved BI dashboard performance by 40% through partitioning and query optimization strategies	
Industrial IoT Predictive Maintenance Platform - Designed end-to-end IoT streaming architecture using Kafka and Spark Structured Streaming for industrial sensor telemetry ingestion - Built predictive maintenance analytics workflows processing high-volume equipment monitoring data in near real time - Developed cloud-native data lake architecture on AWS S3 and Delta Lake supporting time-series analytics	
Real-Time Cybersecurity Threat Detection Pipeline - Built distributed cybersecurity event processing pipeline ingesting over 1M+ security log events daily - Developed anomaly detection workflows for suspicious authentication and network activity analysis - Implemented scalable log analytics architecture using Kafka, Elasticsearch, and Spark Structured Streaming	